

12345678

UNNI C02 SMARTIFICATION – REV A

Two-sheet cleanup: standard power symbols, grid-aligned wiring, explicit local connectivity.

USB + Power

ESP32–C6 + Auxiliary

Simulation Harness

File: 01_usb_power.kicad_sch

File: 02_esp_aux.kicad_sch

File: 03_simulation_harness.kicad_sch

Unni C02 Smartification

Sheet: /
File: unni-smartification-c6.kicad_sch

Title: Unni C02 Smartification – ESP32–C6 Rev A

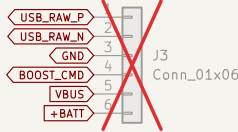
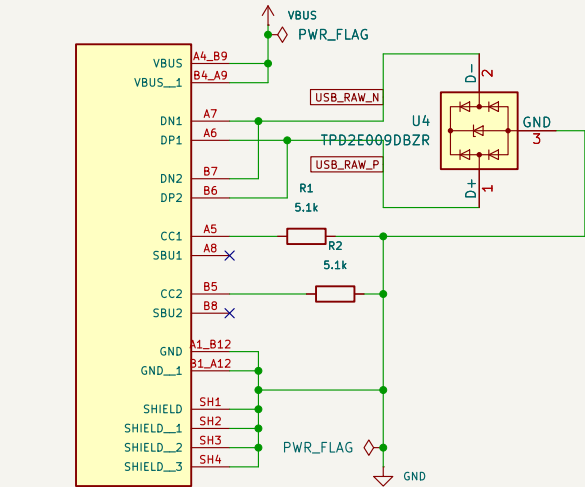
Size: A3Date: 2026–08–19

KiCad E.D.A. 10.99.0–2756–gace8bf0470Id: 1/4

A B C D E F

USB + POWER

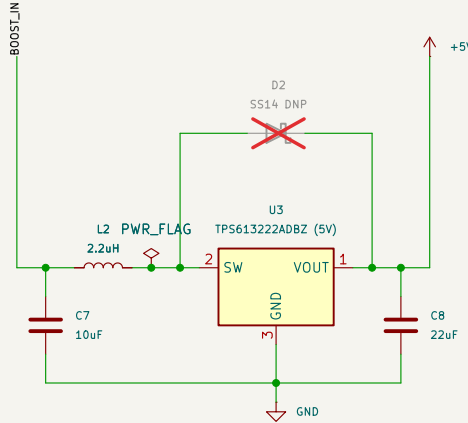
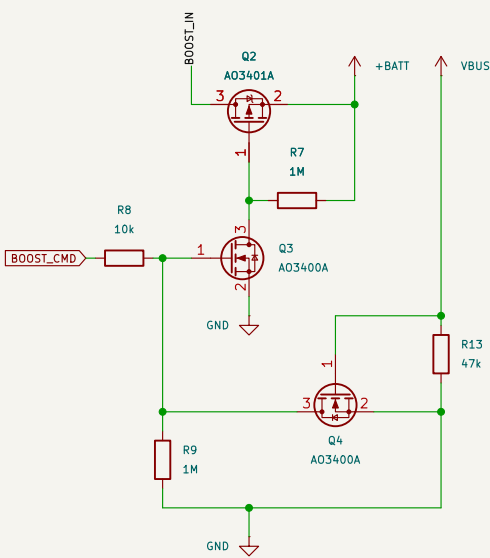
All placement/wiring is on 50 mil (1.27 mm) grid. Standard KiCad symbols used where available.



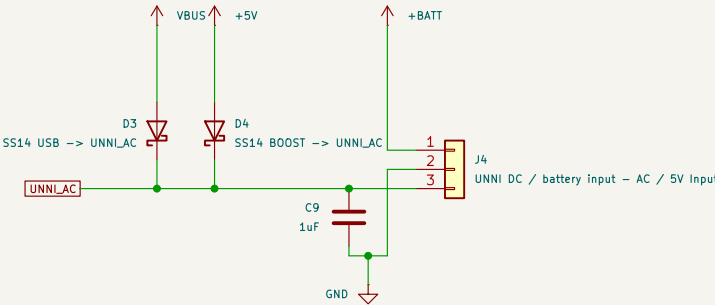
J1
USB-C (replacement daughterboard)

MCP73831 is on the ESP/battery board; R3 = 5.1 kΩ sets ~196 mA; STAT is monitored by the ESP. Battery WTC remains firmware-supervised.

FORCED-AWAKE 5V

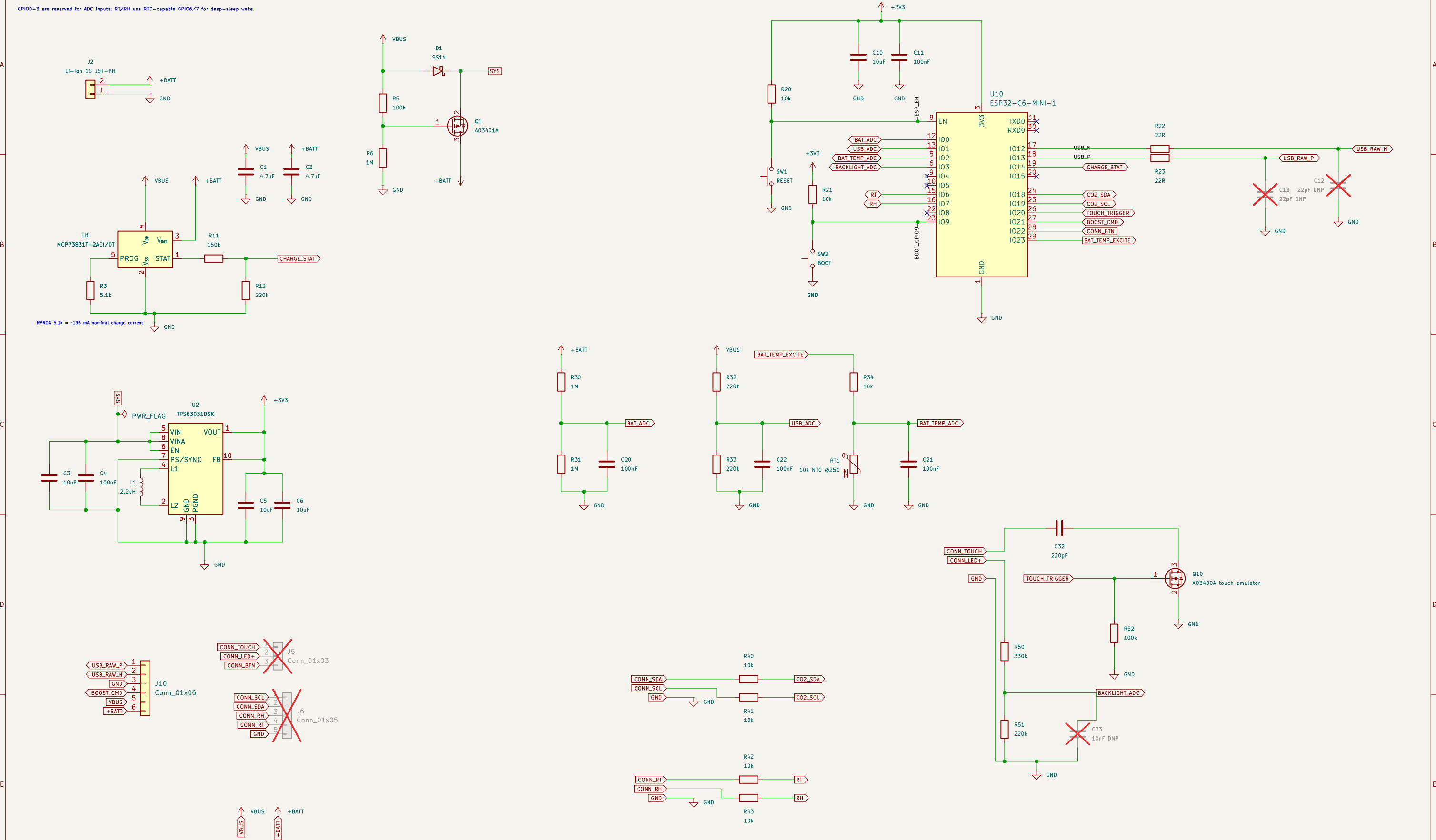


UNNI POWER OUTPUTS



ESP32-C6 + AUXILIARY CIRCUITS

GPIO0-3 are reserved for ADC inputs; RT/RH use RTC-capable GPIO6/7 for deep-sleep wake.



SIMULATION HARNESS – excluded from BOM/PCB

Scenario: battery-only → forced awake → USB hot-plug while BOOST_CMD remains asserted → autonomous charging → USB removal. Battery model includes 120 mΩ ESR. Battery model includes 120 mΩ ESR. I2 is the 3V3-converter input-power proxy. TP5613222A Input power/discharge is modeled Internally; legacy I3 is excluded from simulation.

